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2025

Contents

№4. **,**

: 2

1.

2- , :

$$a_2\ddot{y} + a_1\dot{y} + a_0 y = u. \tag{1}$$

a_2, a_1 a_0 , , :

$$u = k_0 y + k_1 \dot{y} \tag{2}$$

.

:

$$a_2 = 1,$$
$$a_1 = -1,$$
$$a_0 = -2.$$

:

$$\ddot{y} - \dot{y} - 2y = u. \tag{3}$$

:

$$s^2 - s - 2 = 0,$$

$\lambda_1 = 2$ ( ) $\lambda_2 = -1$ ( ).
: $y(0) = 0$, $\dot{y}(0) = 1$.

$u = k_0 y + k_1 \dot{y}$, :

$$\ddot{y} - \dot{y} - 2y = k_0 y + k_1 \dot{y},$$

$$\ddot{y} + (-1 - k_1)\dot{y} + (-2 - k_0)y = 0.$$

:

$$s^2 + (-1 - k_1)s + (-2 - k_0) = 0.$$

:

1. $a_2 = 1 > 0$ — ;
2. $-1 - k_1 > 0 \Rightarrow k_1 < -1$;
3. $-2 - k_0 > 0 \Rightarrow k_0 < -2$.

: $k_0 = -10$, $k_1 = -5$.

:

$$s^2 + 4s + 8 = 0.$$

: $\lambda_{1,2} = -2 \pm 2j$. , , .

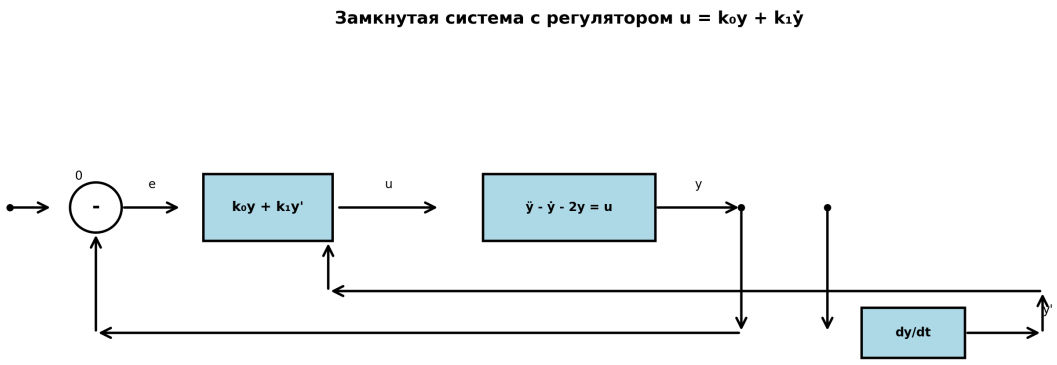


Figure 1: $u = k_0 y + k_1 \dot{y}$

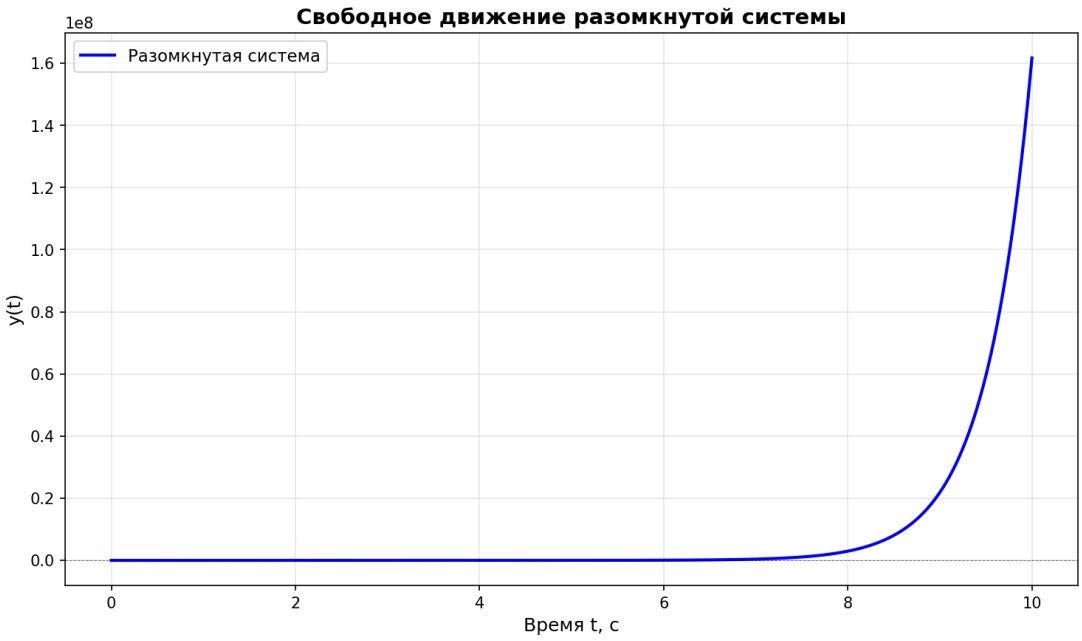


Figure 2:

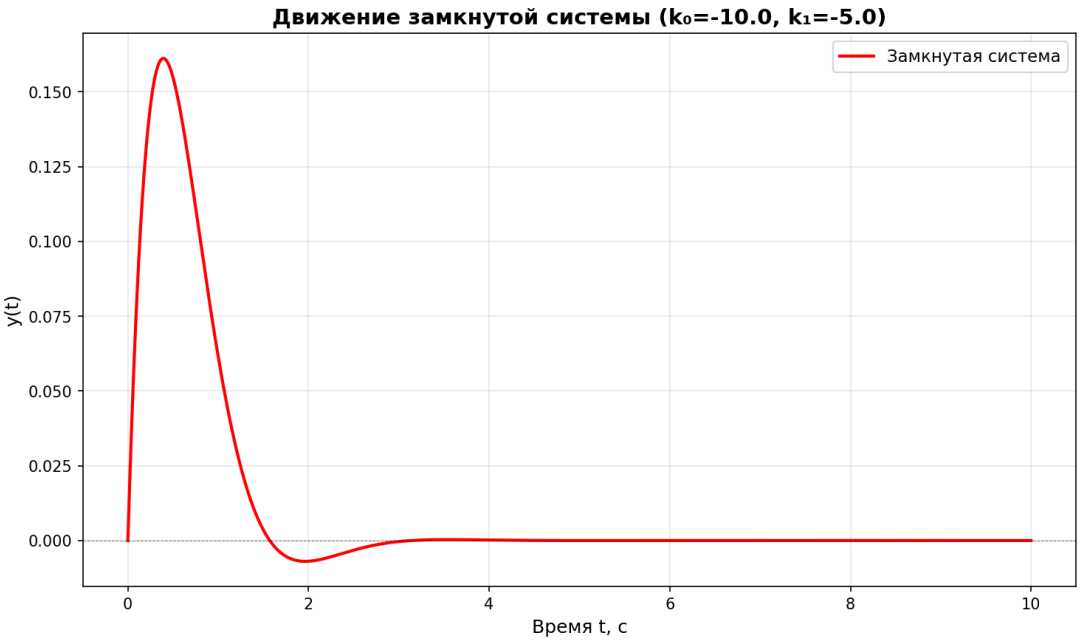


Figure 3:

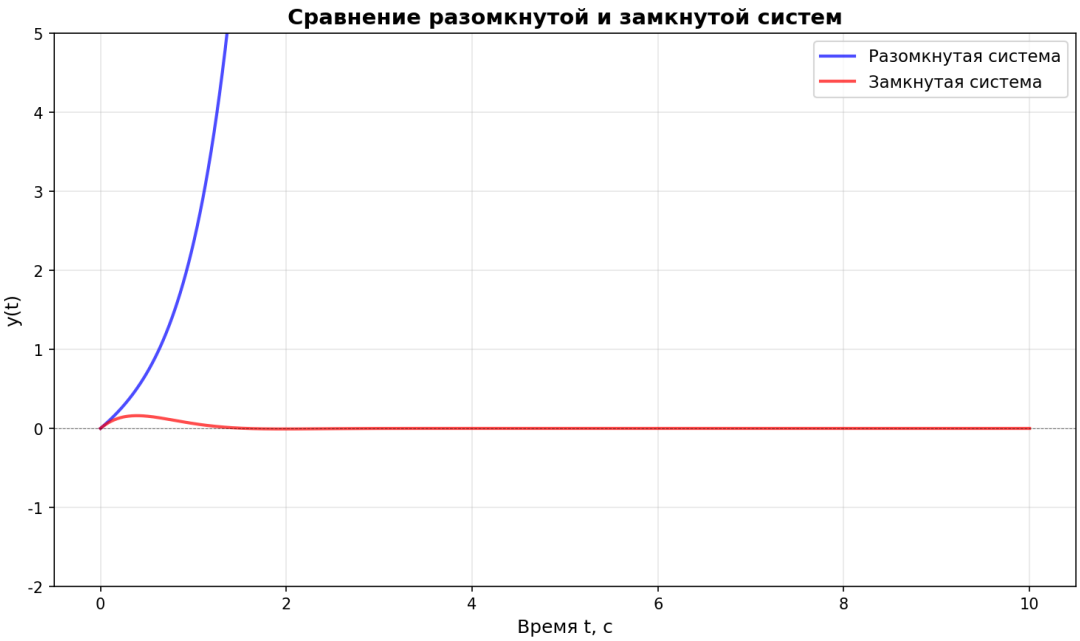


Figure 4:

1

$a_2 = 1,\ a_1 = -1,\ a_0 = -2$,
$\lambda_1 = 2.$.
$u = k_0 y + k_1 \dot{y}$ $k_0 = -10,\ k_1 = -5$.
$\lambda_{1,2} = -2 \pm 2j$, .
, , .

2.

1, :

$$W_{\ldots}(s) = \frac{s}{Ts+1}. \tag{4}$$

T.

. T

. T .

T: 0.01, 0.05, 0.1, 0.2, 0.5.

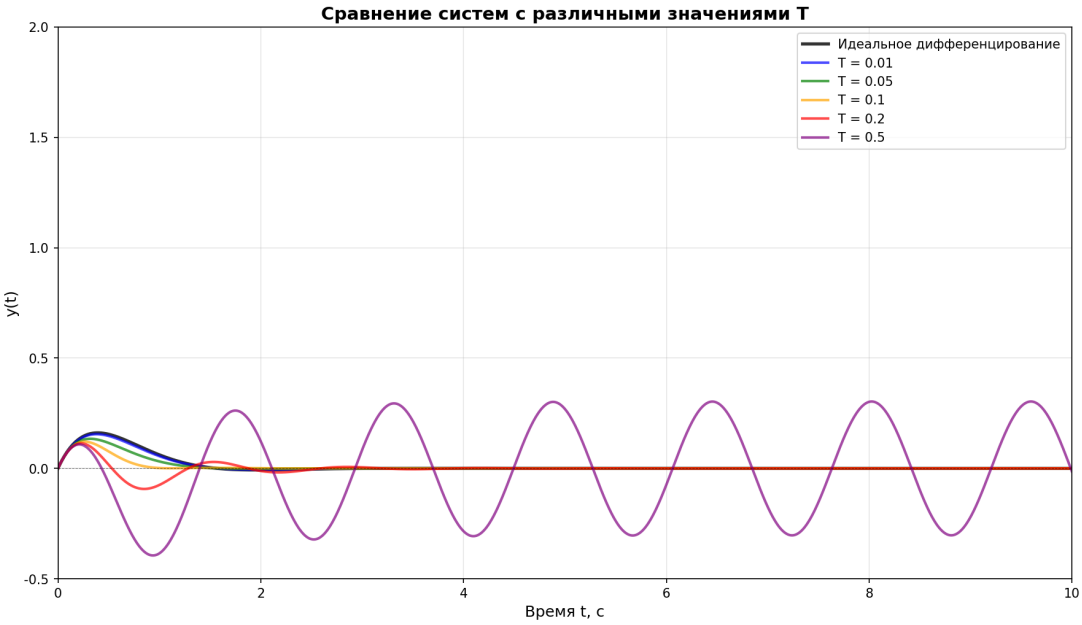
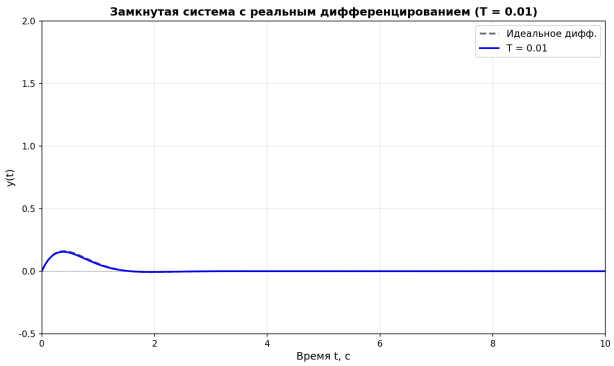
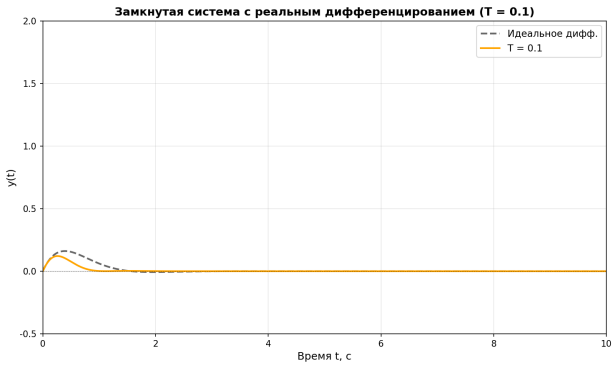


Figure 5: T



(a) T = 0.01



(b) T = 0.10

Figure 6: T

2

, T (0.01, 0.05, 0.1)
. T 0.2 0.5 ,
.

, .
T (0.01-0.05),
.

3. (-)

$H(s) = k$:

$$W(s) = \frac{3}{s^2 + 2.5s + 1}.$$

2 : $A = 2$, $V = 2$.
: $k = 1, 5, 10$.

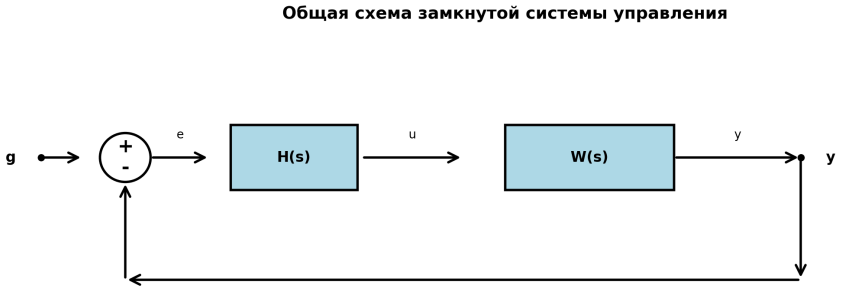


Figure 7:

$g(t) = A = 2$

:

$$e = \lim_{s \to 0} s \cdot \frac{1}{1 + kW(s)} \cdot \frac{A}{s} = \frac{A}{1 + kW(0)} = \frac{A}{1 + 3k}.$$

:

Table 1: k

k	e (.)	e (.)
1	0.5000	0.5000
5	0.1250	0.1250
10	0.0645	0.0645

Figure 8: : y(t) g(t)

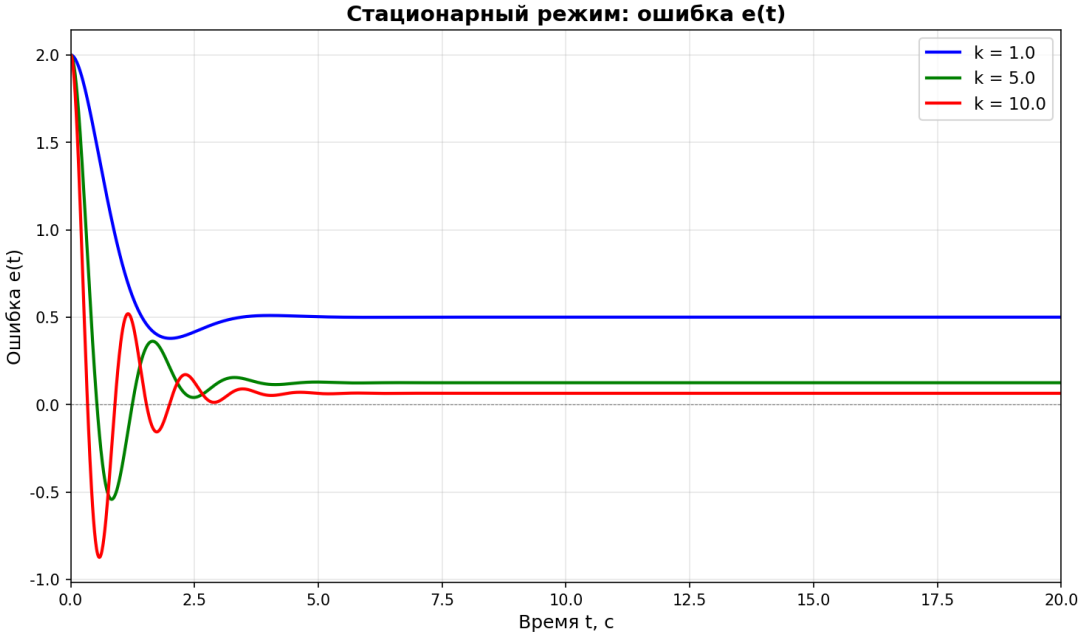


Figure 9: : e(t)

$$g(t) = 2t$$

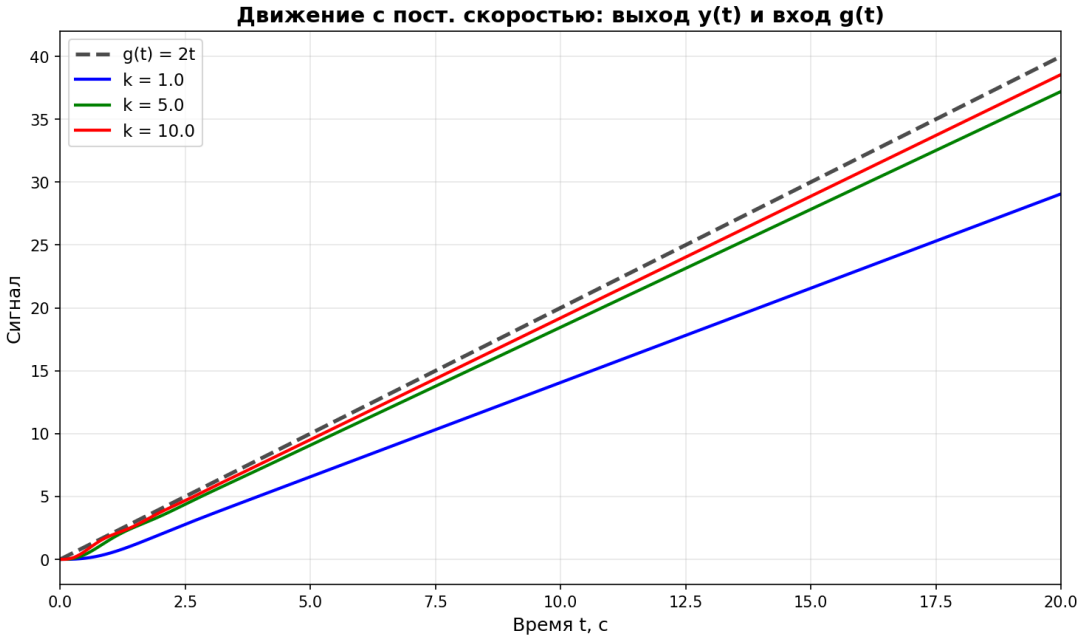


Figure 10: . : y(t) g(t)

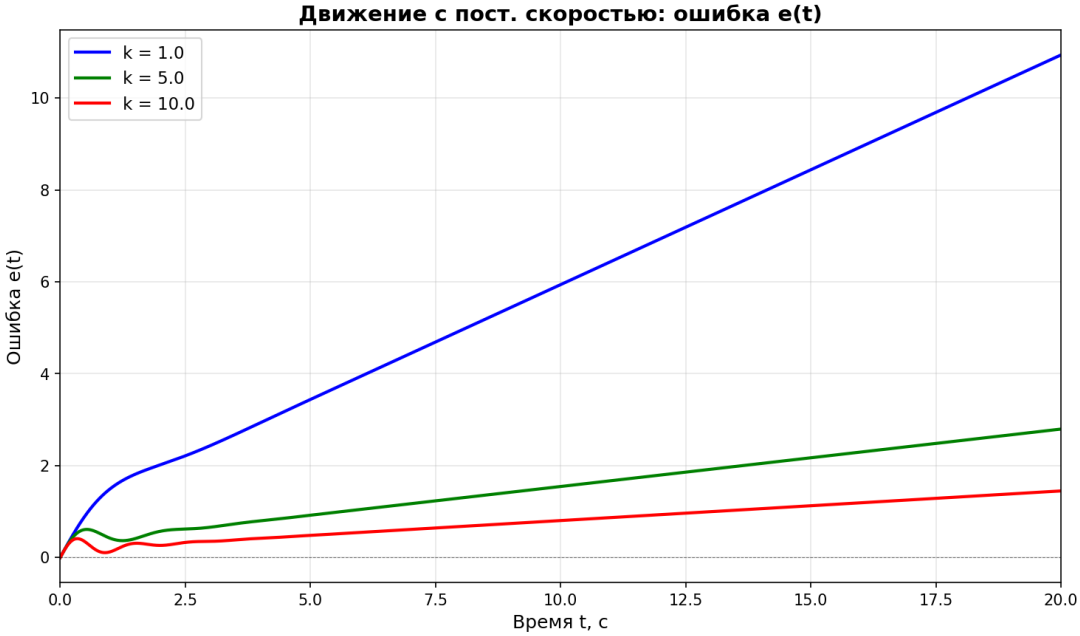


Figure 11: . : e(t)

3

- . ( )
, k. k , .
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.

4. (-)

$H(s) = k/s$ $W(s) = 3/(s^2+2.5s+1)$.
: $k = 0.5, 1, 2$.

$g(t) = 2$

.

Figure 12: - : y(t) g(t)

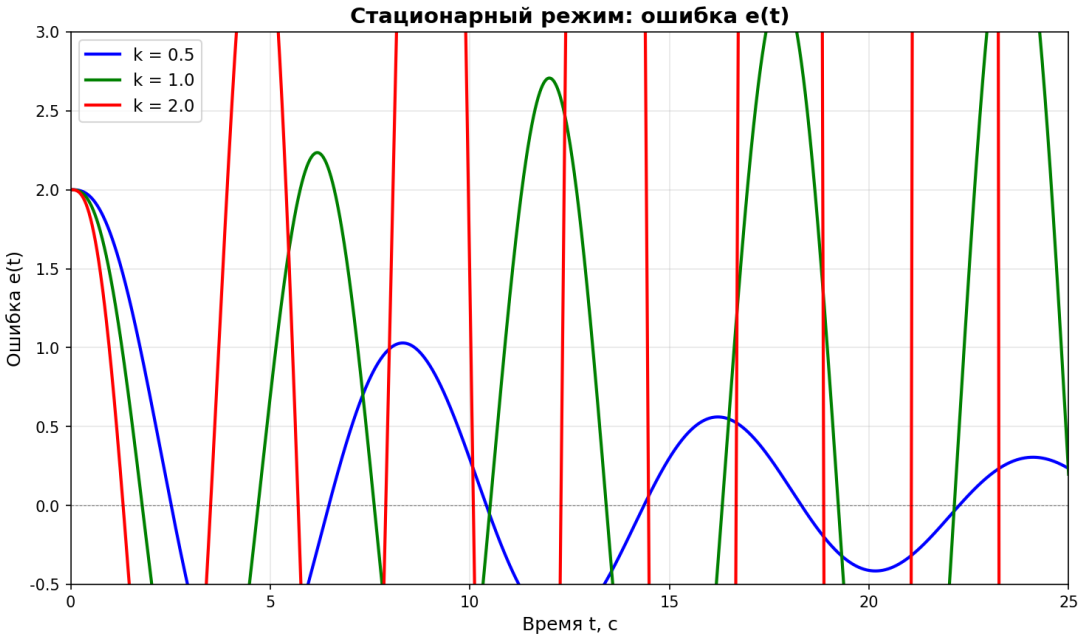

Figure 13: - : e(t)

$$g(t) = 2t$$

.

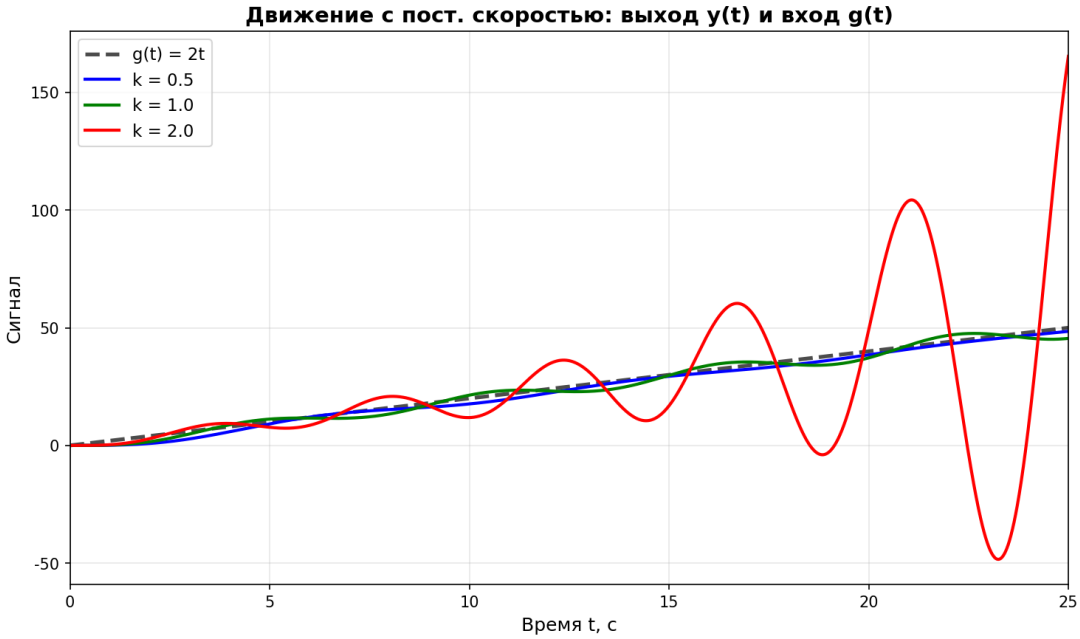

Figure 14: . (-): y(t) g(t)

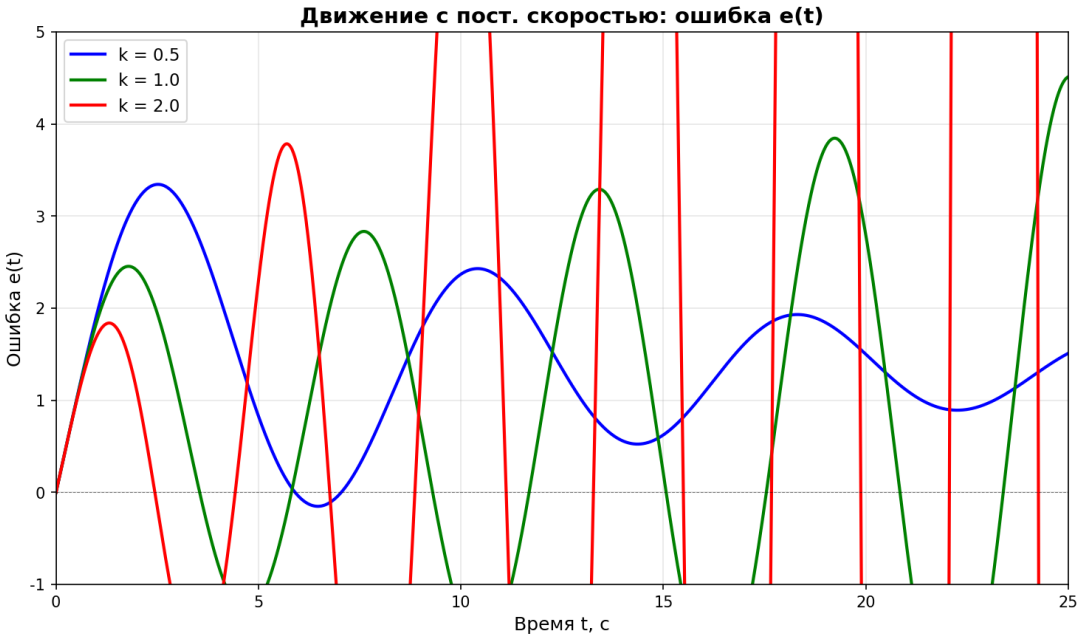

Figure 15: . (-): e(t)

$$g(t) = 0.5t^2$$

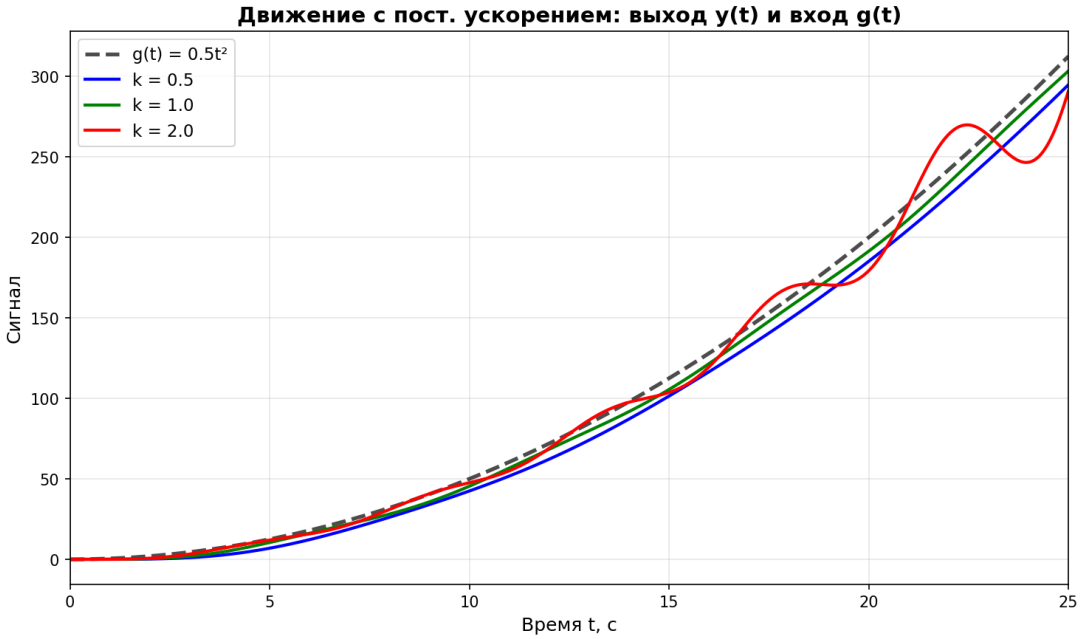

Figure 16: . : y(t) g(t)

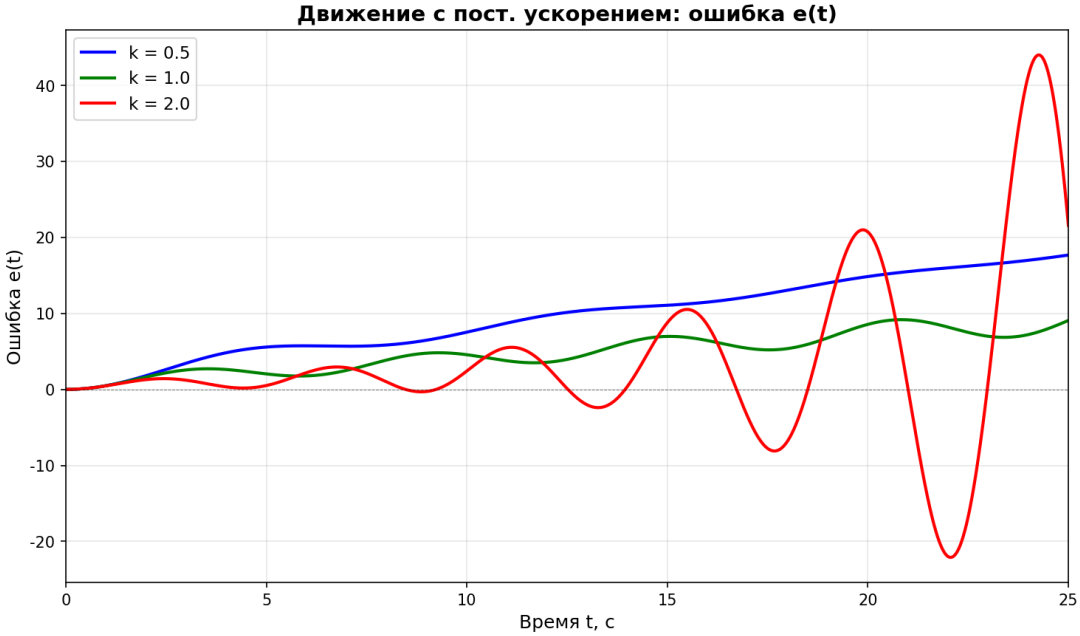

Figure 17: . : e(t)

4

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k , .

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5. (-)

- :

$$H(s) = k\,/s + k\,.$$

:

Table 2: -

	k	k
1	0.5	1.0
2	0.5	2.0
3	1.0	1.0
4	1.0	2.0

$$g(t) = 2t$$

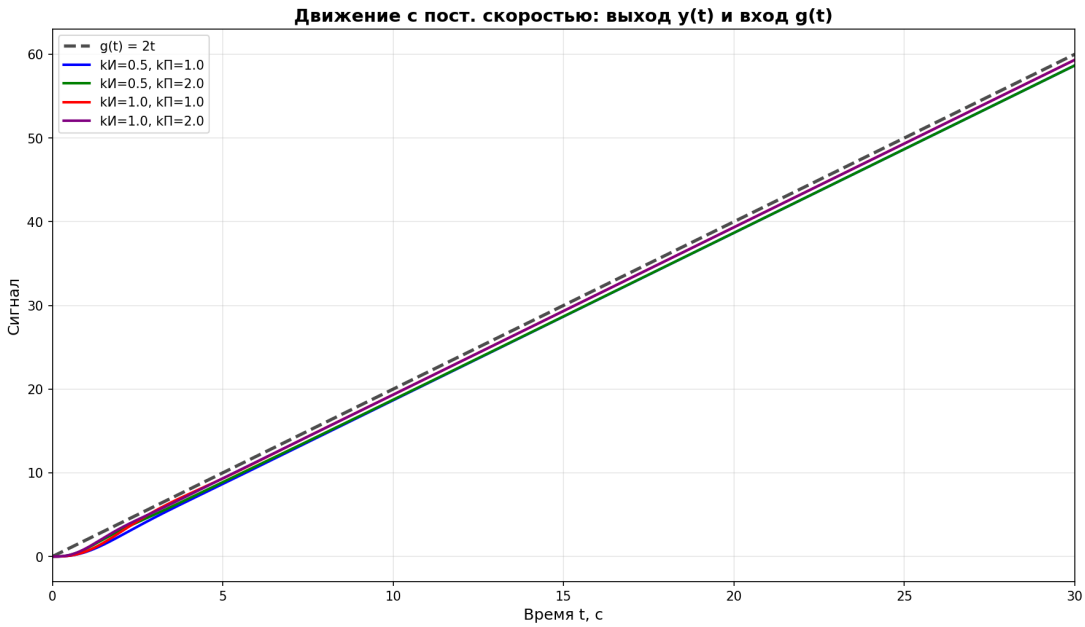

Figure 18: . (-): y(t) g(t)

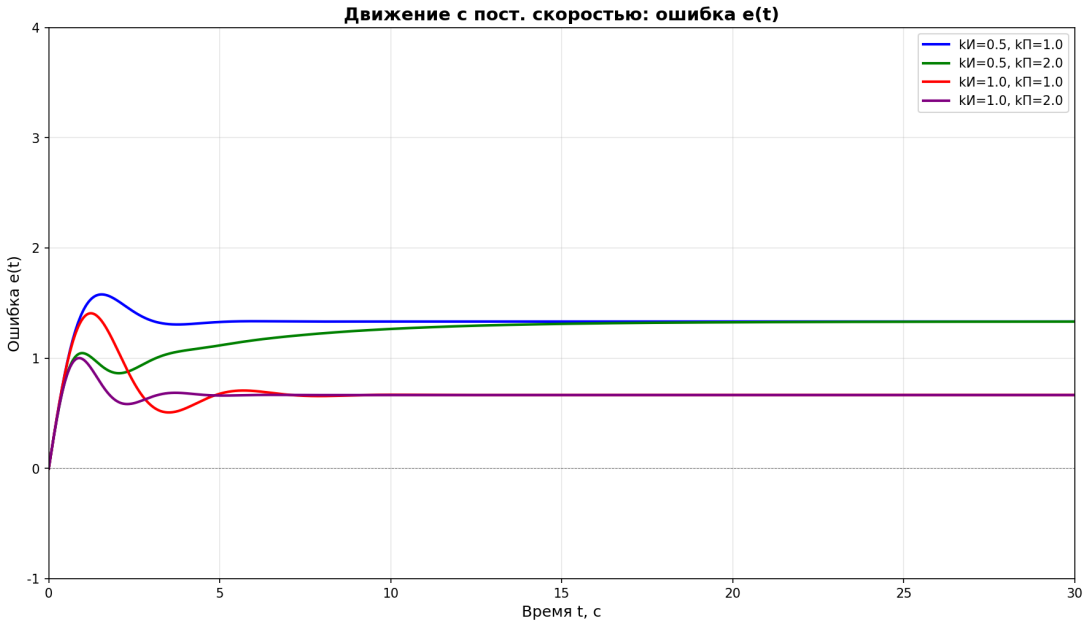

Figure 19: . (-): e(t)

$$g(t) = 2\sin(0.5t)$$

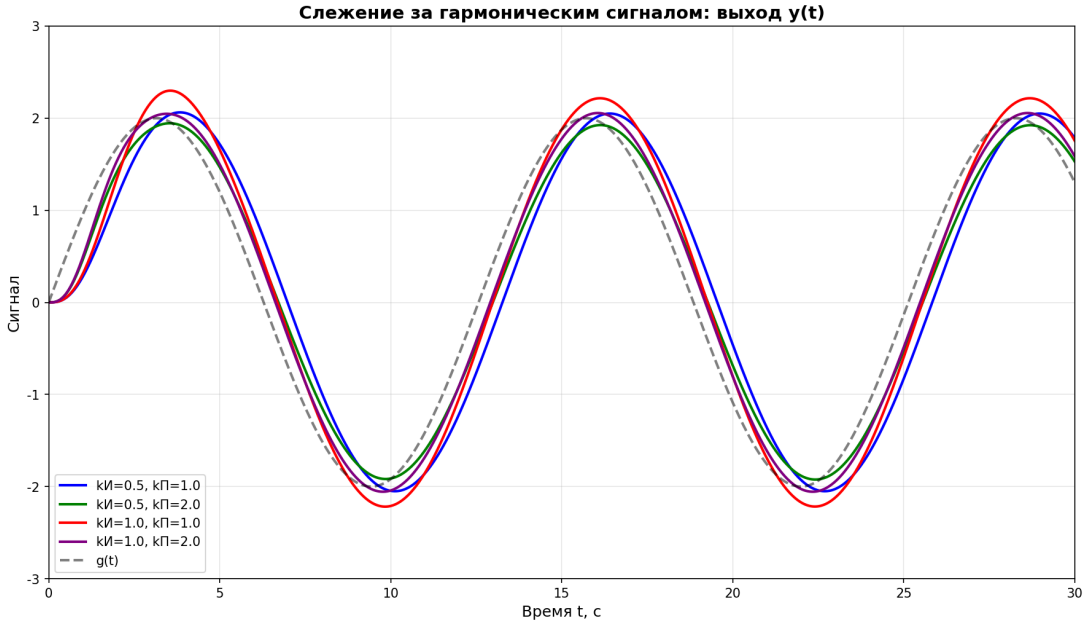

Figure 20: : y(t)

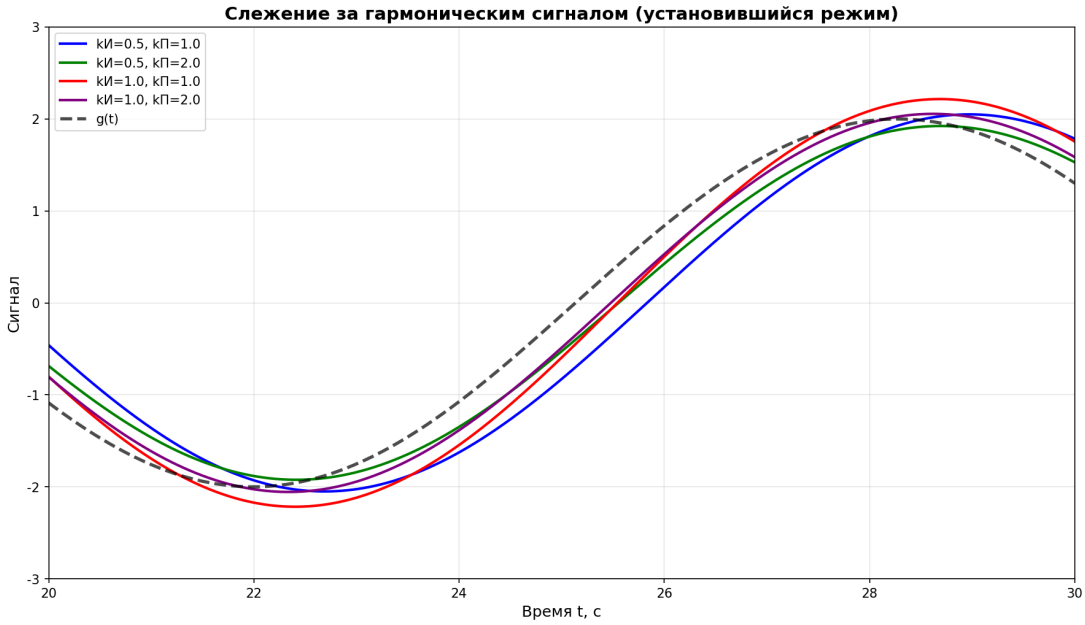

Figure 21: ( )

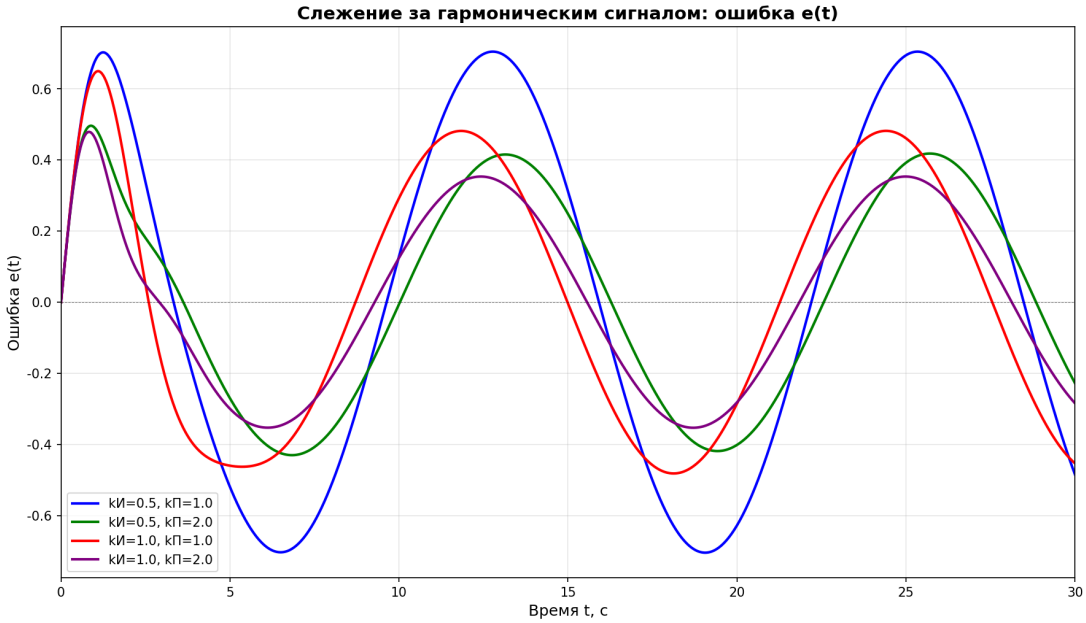

Figure 22:

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. k k , .

6. ( )

$g(t) = 2\sin(0.5t)$:

$$H(s) = \frac{\sum_{k=0}^{m} b_k s^k}{\sum_{k=0}^{m} a_k s^k},$$

.

$e \quad = 0$ $\omega = 0.5$ $s = \pm j\omega = \pm j0.5.$

: $m = 2.$

:

$$M(s) = s^2 + \omega^2 = s^2 + 0.25.$$

4- $\omega_0 = 1$:

$$D(s) = s^4 + 2.6131s^3 + 3.4142s^2 + 2.6131s + 1.$$

$N(s)$:

$$(s^2 + 2.5s + 1)(s^2 + 0.25) + 3N(s) = D(s).$$

:

$$N(s) = 0.0377s^3 + 0.7214s^2 + 0.6627s + 0.25.$$

, :

$$H(s) = \frac{0.0377s^3 + 0.7214s^2 + 0.6627s + 0.25}{s^4 + 10s^3 + 25.25s^2 + 2.5s + 6.25}.$$

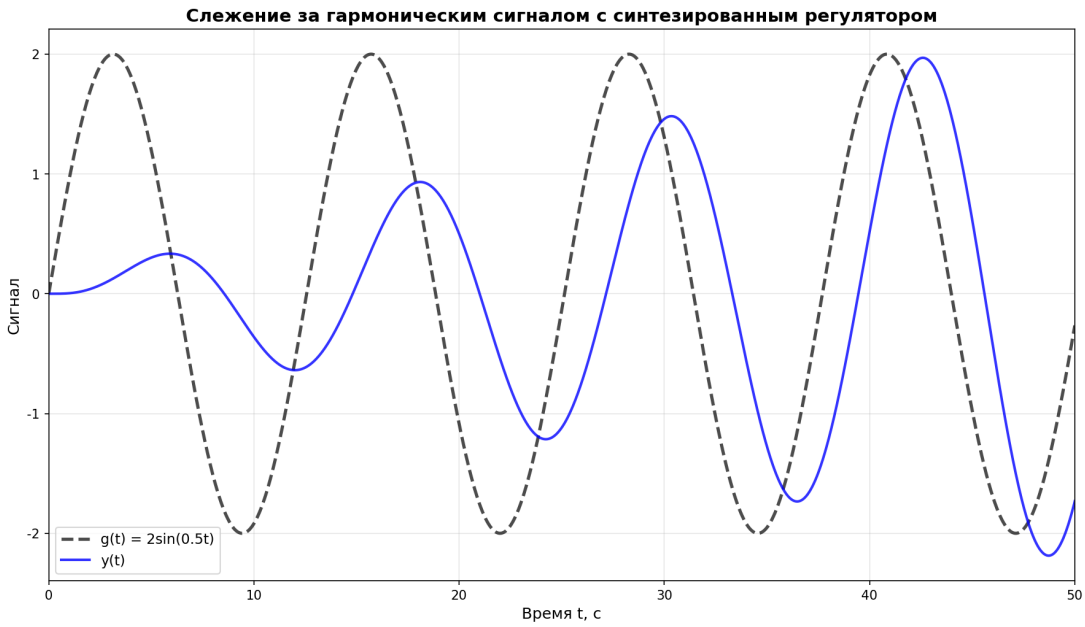

Figure 23:

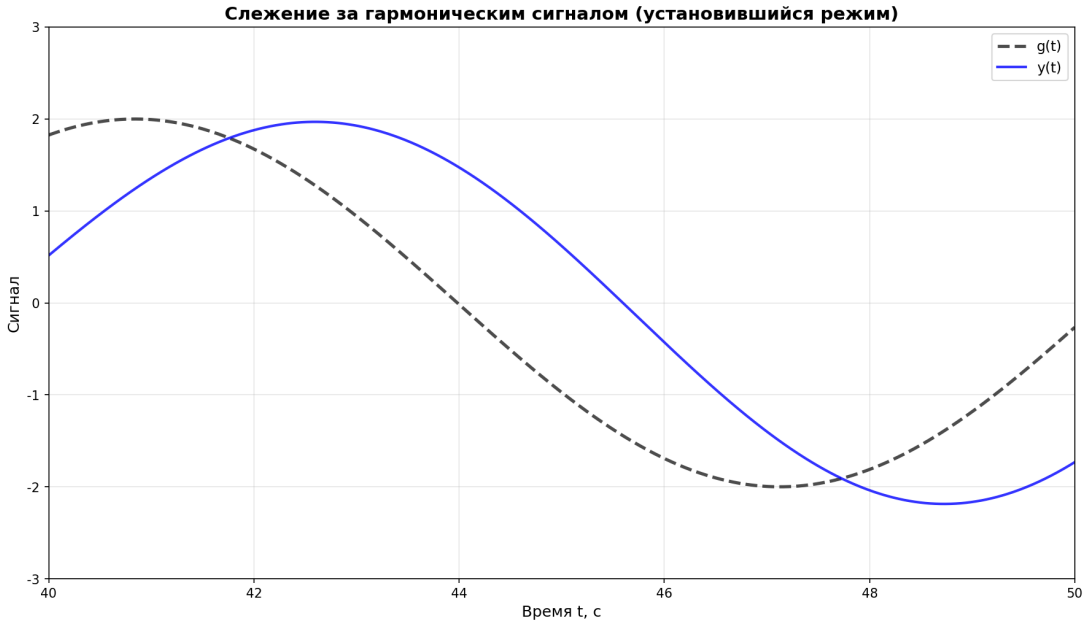

Figure 24: ( )

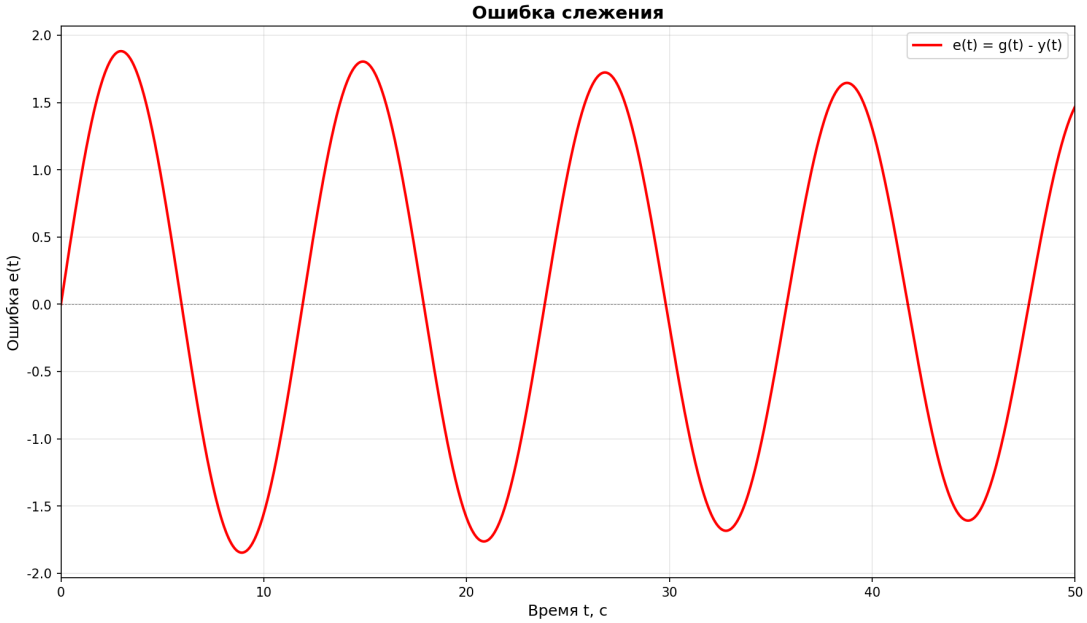

Figure 25:

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$s = \pm j\omega$

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$t \approx 3/\omega_0 = 3$.

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1. **:** ,
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2. **-** (**0-**): ,
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3. **-** (**1-**): , .

4. **-** **:** - - , .

5. **:** ,
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